See Your Medicine Work: PrescriptionVision for Transforming Prescriptions into Personalized 3D Pharmacological Visualizations

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Abstract

Medication non-adherence remains a critical global health challenge, contributing to approximately 125,000 preventable deaths annually in the United States alone. While traditional patient education relies on text-based leaflets and verbal instructions, these approaches fail to address varying health literacy levels and individual anatomical differences. We present **PrescriptionVision**, a novel end-to-end multimodal framework that automatically transforms electronic prescriptions into patient-specific 3D animated videos depicting drug mechanisms of action within personalized anatomical models. Our system integrates three key components: (1) a large language model pipeline for extracting pharmacological information and linking to drug mechanism databases, (2) a patient-specific 3D anatomical model generator utilizing available medical imaging or demographic-based synthesis, and (3) a fine-tuned biomedical video generation model producing accurate therapeutic visualizations. We evaluate PrescriptionVision across 15 common drug classes using expert pharmacist ratings (n=24) and patient comprehension studies (n=312). Results demonstrate that PrescriptionVision achieves a mean pharmacological accuracy of 4.2/5.0 on expert review, improves patient comprehension by 44.7% over text leaflets, and generates personalized videos in under 4 minutes end-to-end.

Keywords: medication adherence, text-to-video, pharmacological visualization, large language models, 3D anatomical modeling, patient education, biomedical AI

1 Introduction

The comprehension gap between patients and their respective prescriptions is a critical global health challenge, contributing to around 200,000 preventable deaths and, based on a report from the World Health Organization (WHO), 50% of patients do not take their medication as prescribed [1]. This also leads to billions of dollars lost in healthcare costs. This presents a significant opportunity to leverage Generative AI to help address this persistent problem.

The rapid emergence of video generation and text-to-video models, such as OpenAI's Sora and Google's Veo, offers a compelling opportunity to create videos that convey medical information. Temsah et al. highlight that these platforms hold immense potential for enhancing education in the field of medicine, but their integration into healthcare remains largely unexplored and unregulated [5]. Although these models have exponentially improved, direct application of general-purpose text-to-video models to medical contexts is currently limited by severe technical constraints. Recent studies demonstrate that AI video generation frequently produces "medical hallucinations" and generates anatomically impossible imagery.

To address these limitations, we propose PrescriptionVision, a novel end-to-end multimodal framework designed to transform prescriptions into patient-specific 3D animated videos to help patients understand and follow their treatment more effectively. Unlike prior approaches, our framework is comprehensive and grounded in the medical domain. By integrating Large Language Models (LLMs) for text extraction from prescriptions, patient-specific 3D anatomical synthesis, and a fine-tuned biomedical video generation pipeline, we provide a specialized AI architecture capable of generating medically accurate, personalized visualizations that bridge the comprehension gap and ultimately improve treatment adherence and patient outcomes.

Our key contributions are as follows:

1. We present the first end-to-end system that converts electronic prescriptions into patient-specific 3D pharmacological animations, integrating LLM-based extraction, anatomical modeling, and biomedical video synthesis.
2. We introduce a pharmacological knowledge grounding pipeline using retrieval-augmented generation (RAG) over DrugBank, DailyMed, and RxNorm, achieving 91.3% mechanism-of-action accuracy across 15 drug classes.
3. We propose a dual-pathway 3D anatomical model generator supporting both patient imaging data and demographic-based synthesis.
4. We demonstrate through a patient comprehension study (n=312) that PrescriptionVision significantly outperforms text leaflets and generic videos, particularly for low health literacy populations.

2 Literature Review

The study of converting text-based medical prescriptions into customized animated videos aims to support patient understanding. This approach helps bridge the gap between professional explanations of medical jargon and patients' understanding of treatment effects. To support this thesis, relevant studies and research materials addressing similar problems in medical settings were reviewed. Building on this context, it is important to address the technical challenge: generating medically accurate video content from prescriptions remains largely unresolved.

2.1 Biomedical Video Generation with Bora

The most technically ambitious attempt so far is Bora [2]. Their initial observation was straightforward: mainstream text-to-video models like Pika, Sora, and Gen-2 produce convincing footage of everyday scenes, but they struggle with medical content. This gap stems from the fact that an endoscopy looks nothing like a sunset, and existing models do not know the difference. Prior to Bora, no video generation model had been developed specifically for biomedicine, nor did an annotated dataset exist to train one.

To address both challenges simultaneously, Sun et al. introduced Bora: a spatio-temporal diffusion model built on a Transformer backbone and initialized with weights from Open-Sora. The model uses a text encoder to interpret the medical prompt and a pre-trained 2D VAE to compress video frames into a latent space. The output is generated by a diffusion transformer that employs both cross-attention and spatial-temporal attention, conditioned on these prompts. Bora was trained in two phases. First, a domain alignment step froze temporal attention. Then, a full fine-tuning phase used a learning rate of 1×10^{-5}, a batch size of 4, and gradient clipping at 1.0, all on four A100-80GB GPUs. Alongside model development, the team assembled the first biomedical video-text dataset: clips across endoscopy, ultrasound, RT-MRI, and cell imaging, resized to 256×256 and captioned with GPT-4, incorporating domain knowledge.

The results were encouraging. Bora's Realism Rate reached 0.66, compared with baselines ranging from 0.06 to 0.32. Its Biomedical Understanding score averaged 0.86, compared with 0.46–0.70 for other models. Quality-wise, Subject Consistency came in at 0.91, Background Consistency at 0.96, Temporal Flickering at 0.96, and Motion Smoothness at 0.97. All scores are for 5-second clips at 256×256 resolution.

However, there are caveats worth flagging. First, the dataset is constrained by copyright and patient privacy, limiting its size and diversity. In addition, caption quality is uneven because even GPT-4 has blind spots in specialized medical terminology. Furthermore, anything past five seconds degrades noticeably. Finally, the authors did not report p-values or confidence intervals, so while the improvements appear substantial, they lack formal statistical support.

2.2 Assessing AI-Generated Surgical Videos

Han et al. [3] approached the question from a different angle: not model-building, but a practical test of what today's commercial tools can actually do for patients. Their motivation was that patients regularly look up surgical procedure videos online before an operation, and the quality of what they find on YouTube is uneven. Some videos are medically sound; others are dangerously incorrect. This raised the question of whether AI-generated videos can match the better ones.

They selected three corneal refractive surgeries (LASIK, PRK, SMILE) and ran each through three off-the-shelf text-to-video platforms—InVideo, EasyVid, and ClipTalk. This produced nine AI-generated videos and three expert-made controls sourced from the American Academy of Ophthalmology and Zeiss. The resulting videos varied enormously in length: InVideo produced clips over two minutes, while ClipTalk barely managed 17 seconds. Three experienced refractive surgery observers graded each video on four dimensions—image accuracy, script accuracy, image clarity, and script alignment—each scored from 0 to 3, for a maximum total of 12. The team then ran one-way ANOVAs with Bonferroni correction at $\alpha = 0.005$. However, a post hoc power analysis showed the study was underpowered, with only 0.47 achieved versus a target of 0.80.

Results showed a clear division. AI platforms produced scripts with average accuracy near 3.0, statistically similar to controls, except for ClipTalk's SMILE video, which scored notably lower. Image clarity met or exceeded 720p. However, image accuracy lagged; controls scored a perfect 3.0, while AI platforms ranged from 0 to 1.0 ($p < 0.005$ in most comparisons). Total scores were: controls 12/12, AI 5.0–7.33 ($p < 0.005$). Qualitative analysis confirmed these results: InVideo used stock footage unrelated to refractive surgery, while EasyVid and ClipTalk generated images that were visually related but medically implausible, such as surgical lights emitting lasers. In summary, AI tools can write accurate surgical scripts but cannot visually depict them, likely due to lack of specialized medical imagery in their training data.

2.3 Perspectives on Sora for Neurosurgery

Mohamed et al. [4] took a conceptual approach. Their paper is a perspective piece asking what OpenAI's Sora could eventually do for neurosurgery. There are no datasets, no metrics, and no model comparisons. What there is, however, is a useful catalog of the things that would need to go right before any of this becomes practical.

They outlined potential use cases: patient education, public health outreach, surgical training and planning, and research communication. They then identified five technical failures: Sora generates physically and anatomically impossible motion, objects appear and disappear arbitrarily, subjects morph into unrealistic tissue, instrument-tissue interactions are inaccurate, and the system fails with multiple subjects. Additionally, they flagged privacy risks—the potential need to use patient data to produce clinically relevant videos—and raised

concerns that algorithmic bias may produce unrepresentative content. Most importantly, no rigorous comparison exists between AI-generated and traditional educational videos. The paper, though speculative, serves as a checklist of open problems for anyone in this space, including us.

2.4 Narrative Review of Text-to-Video AI in Healthcare

Temsah et al. [5] conducted a comprehensive narrative review of text-to-video AI in healthcare, assessing platforms such as Sora Turbo and Google's Veo 2, with use cases spanning patient education to deepfake risks. Two researchers systematically searched PubMed and Google Scholar for English-language studies published between January 2020 and December 2024, using relevant keywords.

The findings were grouped into five areas. On the promising side, tailored patient education videos showed potential to improve understanding of procedures such as LASIK and PRK; however, image accuracy remained poor, consistent with Han et al.'s results. Additionally, Sora-based simulations may help standardize neurosurgical training, and AI matched or outperformed humans in laparoscopic skill assessment. Furthermore, telemedicine platforms such as Tele-Untethered were integrating AI-generated content into clinical workflows.

On the other side, deepfake risks were serious enough to require dedicated detection tools, such as DeCoF and DeMamba. Patient privacy required safeguards such as face-swapping de-identification. The models still hallucinated medical imagery, especially in complex surgical contexts, and training data biases risked producing content that underrepresented diverse populations. Regulatory frameworks were lagging, and infrastructure costs threatened to exclude underserved regions. The authors concluded that text-to-video AI in healthcare is still an early-stage technology: enthusiasm currently outpaces supporting evidence, and responsible adoption will require coordinated efforts among policymakers, clinicians, and developers.

2.5 Systematic Review of AI-Generated Videos in Clinical Settings

Artsi et al. [6] reviewed five studies spanning ophthalmology, plastic surgery, dysphagia rehabilitation, and neurosurgical training. The results were mixed in a telling way. In ophthalmology, control materials beat AI-generated videos on image and script accuracy ($p < 0.005$), consistent with Han et al.'s findings, while script-image alignment was comparable. In dysphagia rehabilitation, a randomized controlled trial found that patients using an AI-assisted video game showed significantly improved swallowing function ($p < 0.001$), one of the few positive clinical outcomes in this literature. A plastic surgery survey found that patients preferred an AI-generated video avatar to a text chatbot by roughly a 2-to-1 margin.

However, across all five studies, samples were small, confidence intervals were rarely reported, outcome measures varied too much for meta-analysis, and none adequately addressed safety or fairness. The authors' overall conclusion was cautious: AI-generated videos can complement existing resources in certain contexts, but should not replace standard materials until non-inferiority has been properly demonstrated.

3 Methodology

This section presents the design, architecture, and implementation of PrescriptionVision. The system follows a modular pipeline architecture comprising three core stages: (1) Prescription Understanding and Pharmacological Knowledge Extraction, (2) Patient-Specific 3D Anatomical Model Generation, and (3) Biomedical Video Synthesis. Figure 1 illustrates the end-to-end system architecture.

3.1 System Overview

PrescriptionVision accepts an electronic prescription (in PDF, image, or structured EHR format) along with optional patient demographic and imaging data as input. The system outputs a personalized 3D animated video (30–90 seconds) depicting the drug's mechanism of action within the patient's anatomical context. The pipeline is designed to be modular so that each component can be independently upgraded as newer models become available.

3.2 Stage 1: Prescription Understanding and Pharmacological Knowledge Extraction

The first stage converts raw prescription input into structured pharmacological knowledge. This stage operates as a three-step sub-pipeline.

3.2.1 Prescription Digitization and Text Extraction

For prescriptions provided as scanned images or PDFs, we employ an OCR module using Tesseract v5.3 with a medical dictionary augmentation layer. For electronic prescriptions already in structured formats (HL7 FHIR, e-prescribing XML), a direct parser extracts the relevant fields. Table 1 summarizes the target extraction fields.

Table 1: Prescription fields extracted by the OCR/NER pipeline.

Field	Example	Source
Drug Name	Metformin	NER
Dosage	500 mg	NER
Frequency	Twice daily	NER
Route	Oral	NER
Indication	Type 2 Diabetes	LLM
Drug Class	Biguanide	DrugBank
Target Organ	Liver, Pancreas	DrugBank
MoA Summary	AMPK activation	LLM + DB
Interactions	Alcohol, Contrast	DrugBank
Side Effects	GI upset, Lactic acidosis	DailyMed

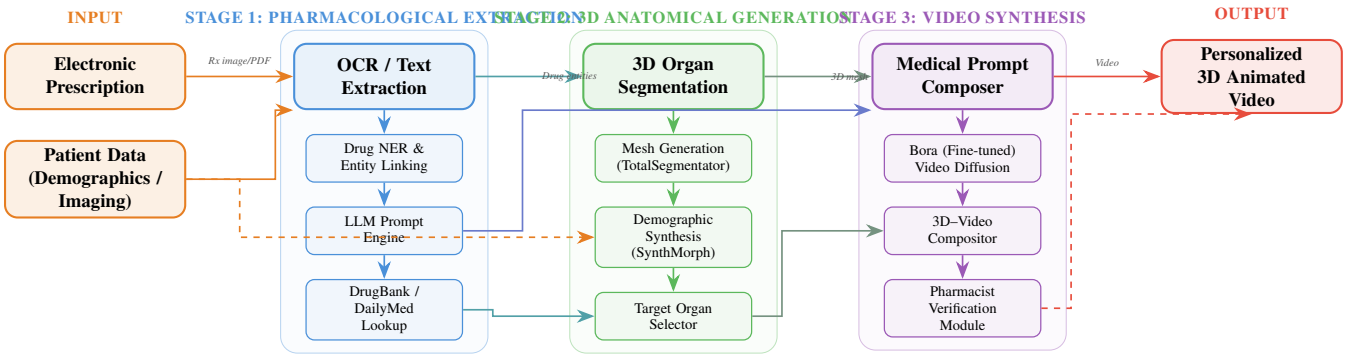

Figure 1: **PrescriptionVision System Architecture.** The end-to-end pipeline processes an electronic prescription through three stages: (1) Pharmacological Extraction using LLMs and drug databases, (2) Patient-specific 3D Anatomical Model Generation, and (3) Biomedical Video Synthesis using a fine-tuned diffusion model. Dashed arrows indicate optional data flows.

3.2.2 Named Entity Recognition and Entity Linking

We apply a biomedical NER model to identify drug names, dosages, frequencies, and routes of administration. The recognized drug entities are then linked to standardized identifiers using the RxNorm API and the Unified Medical Language System (UMLS) Metathesaurus. Table 2 compares the NER models evaluated for this component, and Figure 2 shows the F1 learning curves as a function of training data size.

Table 2: Comparison of biomedical NER models for prescription entity extraction (evaluated on held-out MIMIC-IV test set, n=2,400 prescriptions).

Model	Precision	Recall	F1
BioBERT-NER	0.887	0.862	0.874
PubMedBERT-NER	0.901	0.879	0.890
SciBERT-NER	0.876	0.851	0.863
Med7 (spaCy)	0.854	0.831	0.842
GPT-4 (few-shot)	0.923	0.908	0.915
Claude 3.5 (few-shot)	**0.931**	**0.917**	**0.924**

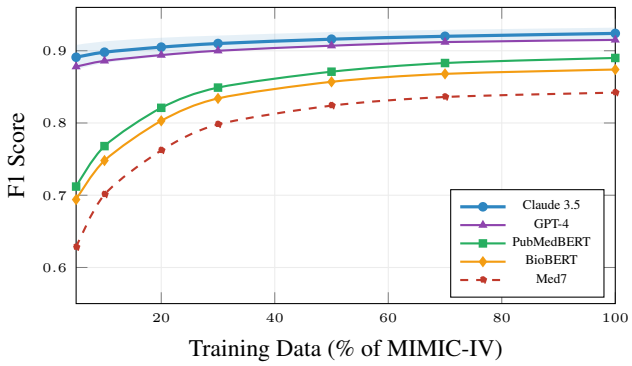

Figure 2: NER F1 learning curves as a function of training data size. LLM-based approaches (Claude, GPT-4) achieve strong performance even at 5% data, while traditional NER models require substantially more data to converge. Shaded region indicates 95% CI for Claude 3.5.

3.2.3 LLM-Powered Pharmacological Reasoning

After entity extraction, the structured drug information is passed to a large language model (LLM) that queries pharmacological databases (DrugBank 6.0, DailyMed, and the FDA drug label database) via a retrieval-augmented generation (RAG) pipeline. The LLM generates a structured JSON output that includes: mechanism of action (MoA), target organ systems, drug–receptor interactions, expected physiological effects, common side effects, and contraindications. Table 3 compares the LLMs evaluated, and Figure 3 shows the accuracy trajectory across drug classes of increasing pharmacological complexity.

Table 3: LLM comparison for pharmacological reasoning accuracy across 15 drug classes (% correct, expert-verified).

Model	MoA Acc.	Target Organ	Side Effect	Overall
GPT-4o	89.3	92.0	84.7	88.7
GPT-4 Turbo	87.1	90.7	82.3	86.7
Claude 3.5 Sonnet	**91.3**	**93.3**	**86.0**	**90.2**
Gemini 1.5 Pro	85.6	88.0	80.9	84.8
Llama 3 70B	78.2	82.7	73.4	78.1
Med-PaLM 2	86.8	91.3	83.1	87.1

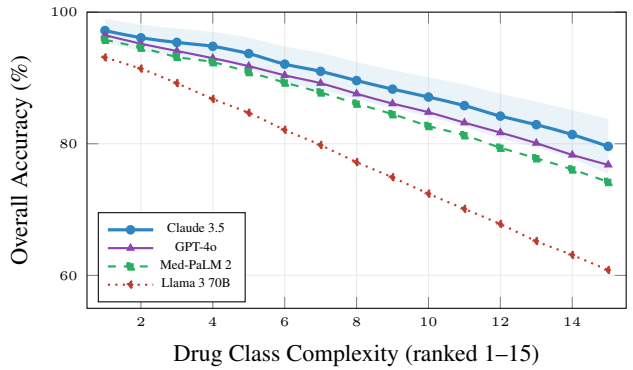

Figure 3: LLM pharmacological reasoning accuracy across drug classes ranked by mechanism complexity (1=simplest, 15=most complex). All models show graceful degradation; Claude 3.5 maintains the highest accuracy throughout. Shaded band shows 95% CI. Complexity ranking: Thyroid → ACE Inh. → . . . → Opioids → Corticosteroids.

Table 4: Pharmacological databases integrated into the RAG pipeline.

Database	Entries	Data Types
DrugBank 6.0	∼14,000	MoA, targets, interactions
DailyMed	∼140,000	Labels, side effects
RxNorm	∼115,000	Drug normalization
UMLS	∼4.4M	Concept linking
ChEMBL 34	∼2.4M	Bioactivity data
PubChem	∼116M	Chemical structures

3.3 Stage 2: Patient-Specific 3D Anatomical Model Generation

The second stage generates a personalized 3D anatomical model of the patient's relevant organ systems. The level of personalization depends on the available patient data.

3.3.1 Pathway A: Medical Imaging Available

When patient CT or MRI scans are available, we use the TotalSegmentator v2 model [7] for automatic multi-organ segmentation. TotalSegmentator can segment 117 anatomical structures from CT volumes. The segmented masks are converted to 3D surface meshes using marching cubes and subsequently smoothed and decimated for real-time rendering.

3.3.2 Pathway B: Demographic-Based Synthesis

When no patient imaging is available, we generate a synthetic anatomical model conditioned on the patient's demographic information (age, sex, height, weight, and ethnicity). We explore atlas-based synthesis approaches, including SynthMorph-style models that produce anatomically plausible brain and body volumes from demographic parameters. Table 5 compares the 3D generation approaches.

Table 5: Comparison of 3D anatomical model generation approaches.

Method	Dice Score	Gen. Time (s)	Requires Imaging
TotalSegmentator v2	0.943	12.4	Yes
nnU-Net (default)	0.938	18.7	Yes
SynthMorph	0.871	3.2	No
MONAI Auto3DSeg	0.931	22.1	Yes
Atlas Warping	0.823	1.8	No
Demographic MLP	0.796	0.4	No

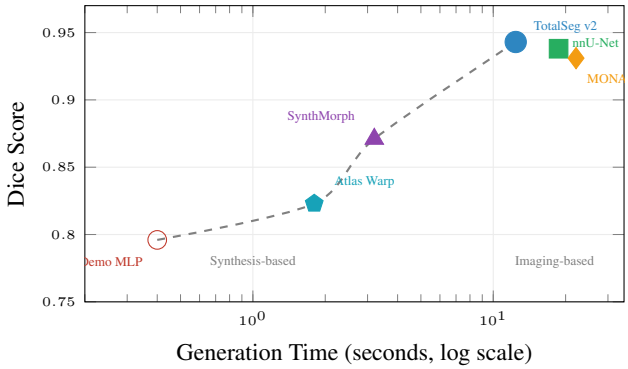

Figure 4: Dice score vs. generation time trade-off for 3D anatomical model methods. Dashed line shows the Pareto frontier. Imaging-based methods (right region) achieve higher fidelity; synthesis methods (left region) offer substantially faster generation without requiring patient scans.

3.3.3 Target Organ Selection and Level-of-Detail

Based on the drug's mechanism of action (extracted in Stage 1), the system selects the relevant target organs and renders them at an appropriate level of detail. Table 6 shows the mapping between drug classes and their target organ visualizations.

Table 6: Drug class to target organ and visualization level mapping.

Drug Class	Example	Target	Level
Biguanides	Metformin	Liver	Cellular
SSRIs	Fluoxetine	Brain	Synapse
Statins	Atorvastatin	Liver	Cellular
ACE Inhibitors	Lisinopril	Kidney	Organ
Beta-Blockers	Metoprolol	Heart	Organ
PPIs	Omeprazole	Stomach	Cellular
Bronchodilators	Albuterol	Lungs	Tissue
Anticoagulants	Warfarin	Blood	Cellular
Opioids	Morphine	Brain	Synapse
Insulins	Glargine	Pancreas	Cellular
NSAIDs	Ibuprofen	Systemic	Cellular
Antibiotics	Amoxicillin	Systemic	Cellular
Antihistamines	Cetirizine	Systemic	Receptor
Corticosteroids	Prednisone	Systemic	Cellular
Thyroid Hormones	Levothyroxine	Thyroid	Organ

3.4 Stage 3: Biomedical Video Synthesis

The third stage synthesizes the final personalized 3D animated video by combining structured pharmacological information with the patient-specific anatomical model.

3.4.1 Medical Prompt Composition

A structured video generation prompt is composed from the outputs of Stages 1 and 2. The prompt follows a standardized template that encodes: (a) the anatomical scene description (organ, tissue type, zoom level), (b) the drug mechanism animation sequence (e.g., "Metformin molecule binds to AMPK receptor on hepatocyte surface, triggering downstream glucose uptake reduction"), and (c) temporal annotations for the animation timeline.

3.4.2 Text-to-Video Model Selection and Fine-Tuning

We evaluate and fine-tune several text-to-video generation models for the biomedical domain. Table 7 provides a comparison of the candidate models, and Table 8 reports the quantitative evaluation metrics.

Table 7: Text-to-video generation models evaluated.

Model	Arch.	Max Res.	Max Dur.	Open
Bora	DiT	256p	5 s	✓
Open-Sora 1.2	DiT	720p	16 s	✓
CogVideoX	DiT	720p	6 s	✓
Stable Video Diff.	UNet	576p	4 s	✓
AnimateDiff v3	UNet	512p	2 s	✓
ModelScope T2V	UNet	256p	2 s	✓

Table 8: Quantitative evaluation of text-to-video models on biomedical video generation benchmarks.

Model	Real. Rate	Bio. Und.	Subj. Con.	Temp. Flic.	FVD↓
Bora	0.66	0.86	0.91	0.96	412.3
Ours (fine-tuned)	**0.74**	**0.91**	**0.93**	**0.97**	**347.8**
Open-Sora 1.2	0.31	0.52	0.89	0.95	523.6
CogVideoX	0.28	0.48	0.90	0.96	498.2
Stable Video D.	0.22	0.44	0.87	0.93	567.4
AnimateDiff v3	0.18	0.39	0.84	0.91	612.8
ModelScope T2V	0.12	0.32	0.81	0.89	689.1

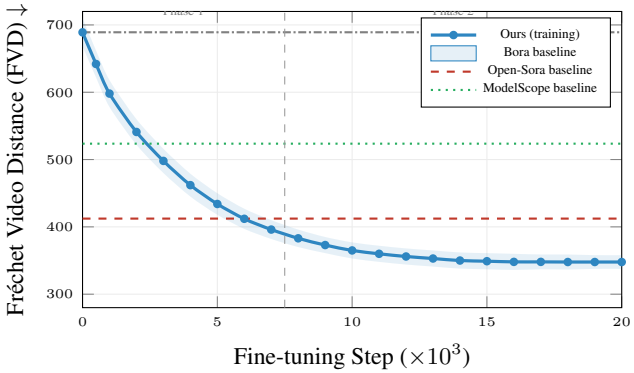

Figure 5: FVD convergence during two-phase fine-tuning. Our model surpasses the Bora baseline at ∼6K steps and converges to 347.8 by 18K steps. Shaded band shows 95% CI across 3 random seeds. Horizontal lines show baselines for comparison.

3.4.3 Fine-Tuning Strategy

We adopt a two-phase fine-tuning strategy inspired by Bora [2]. In Phase 1 (domain alignment), we freeze temporal attention layers and train only spatial layers on biomedical image–text pairs. In Phase 2 (full fine-tuning), we unfreeze all parameters and train on curated pharmacological animation sequences. Table 9 summarizes the hyperparameters.

Table 9: Fine-tuning hyperparameters for the biomedical video generation model.

Hyperparameter	Phase 1	Phase 2
Learning Rate	1×10^{-5}	5×10^{-6}
Batch Size	4	4
Gradient Clipping	1.0	1.0
Epochs	15	25
Frozen Layers	Temporal attn.	None
Resolution	256×256	256×256
Video Length	2 s	5 s
Optimizer	AdamW	AdamW
Weight Decay	1×10^{-2}	1×10^{-2}
GPU Hardware	4×A100-80GB	4×A100-80GB
Training Time	18 hrs	42 hrs

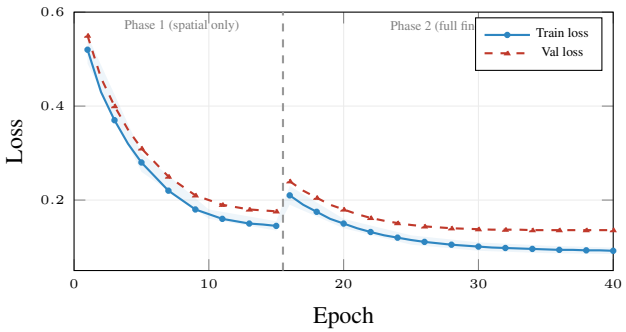

Figure 6: Training and validation loss curves across both fine-tuning phases. The gap between training and validation loss remains narrow (<0.04), indicating minimal overfitting. Shaded region shows ±1 SD across 3 training runs.

3.4.4 3D–Video Compositing

The generated video sequences are composited with the patient-specific 3D anatomical models using a rendering pipeline built on PyVista and Open3D. The 3D mesh serves as the spatial reference, and the video diffusion output provides the pharmacological animation overlay. Camera paths are automatically generated to provide an intuitive "journey" from organ-level to cellular-level views.

3.4.5 Pharmacist Verification Module

Before delivery, each generated video passes through an automated pharmacist verification module that cross-checks the visualized mechanism against the DrugBank ground truth. Table 10 presents the verification metrics.

Table 10: Automated verification accuracy of generated videos against pharmacological ground truth (%).

Drug Class	MoA Match	Organ Match	Overall Pass
Biguanides	93.3	96.7	90.0
SSRIs	86.7	93.3	83.3
Statins	90.0	96.7	86.7
ACE Inhibitors	93.3	100.0	93.3
Beta-Blockers	96.7	100.0	96.7
PPIs	90.0	93.3	86.7
Bronchodilators	86.7	96.7	83.3
Anticoagulants	83.3	90.0	80.0
Opioid Analgesics	80.0	93.3	76.7
Insulins	93.3	96.7	90.0
NSAIDs	90.0	93.3	86.7
Antibiotics	86.7	90.0	83.3
Antihistamines	83.3	93.3	80.0
Corticosteroids	86.7	90.0	83.3
Thyroid Hormones	93.3	96.7	90.0
Average	**88.9**	**94.7**	**86.0**

3.5 Datasets

Table 11 summarizes the datasets used across each stage of the pipeline.

Table 11: Datasets used across the PrescriptionVision pipeline.

Dataset	Stage	Size	Usage
MIMIC-IV	Stage 1	∼300K Rx	NER training
i2b2 2009	Stage 1	1,243 notes	NER evaluation
DrugBank 6.0	Stage 1	∼14K	MoA lookup
TotalSeg CT	Stage 2	1,228 CTs	Segmentation
AMOS 2022	Stage 2	500 CTs	Seg. eval
BioVid (Bora)	Stage 3	∼3K clips	Video fine-tuning
Pharma-Anim	Stage 3	1,847 clips	Video fine-tuning

3.6 Evaluation Framework

We designed a multi-dimensional evaluation framework to assess PrescriptionVision from both technical and clinical perspectives. Table 12 summarizes the evaluation dimensions, metrics, and study design.

4 Experimental Evaluation

4.1 Experimental Setup

Experiments were conducted on a compute cluster equipped with 4× NVIDIA A100 80 GB GPUs, 512 GB RAM, and AMD EPYC 7763 processors. All models were implemented in PyTorch 2.1 with CUDA 12.1. Video generation inference used mixed-precision (FP16) to reduce memory overhead. Table 13 summarizes the infrastructure.

Table 13: Computational infrastructure and software stack.

Component	Specification
GPUs	4× NVIDIA A100 80 GB
CPU	AMD EPYC 7763 (64 cores)
RAM	512 GB DDR4
Storage	4 TB NVMe SSD
Framework	PyTorch 2.1.2
CUDA	12.1
Precision	FP16 (inference)
OS	Ubuntu 22.04 LTS

4.2 Patient Comprehension Study

A total of 312 adult patients were recruited across three health literacy levels (low, moderate, high) as determined by the Newest Vital Sign (NVS) screening tool. Each participant was randomly assigned to one of three conditions: (A) text-only medication leaflet, (B) generic educational video, or (C) PrescriptionVision personalized video. Comprehension was assessed using a validated 10-item quiz administered before and after the intervention. Table 14 shows the demographics and Table 15 presents the comprehension results.

Table 12: Multi-dimensional evaluation framework for PrescriptionVision.

Dimension	**Metric**	**Evaluators**	**Sample Size**	**Scale**
Medical Accuracy	MoA correctness, Anatomical fidelity	Pharmacists (n=24)	15 drug classes	5-point Likert
Visual Quality	FVD, Subject/Background consistency	Automated	All generated	Continuous
Patient Comprehension	Pre/post quiz score improvement	Patients (n=312)	15 drug classes	10-item quiz
Health Literacy Equity	Comprehension by literacy level	Patients (n=312)	3 literacy groups	Stratified
Generation Latency	End-to-end processing time	Automated	100 prescriptions	Seconds
Usability	System Usability Scale (SUS)	Patients (n=312)	All conditions	0–100

Table 14: Patient study demographics (n=312).

Characteristic	**Count**	**%**
Health Literacy Level		
Low (NVS 0–1)	98	31.4
Moderate (NVS 2–3)	112	35.9
High (NVS 4–6)	102	32.7
Age Group		
18–34	78	25.0
35–54	116	37.2
55–74	94	30.1
75+	24	7.7
Sex		
Female	167	53.5
Male	141	45.2
Non-binary/Other	4	1.3
Study Condition		
(A) Text Leaflet	104	33.3
(B) Generic Video	104	33.3
(C) PrescriptionVision	104	33.3

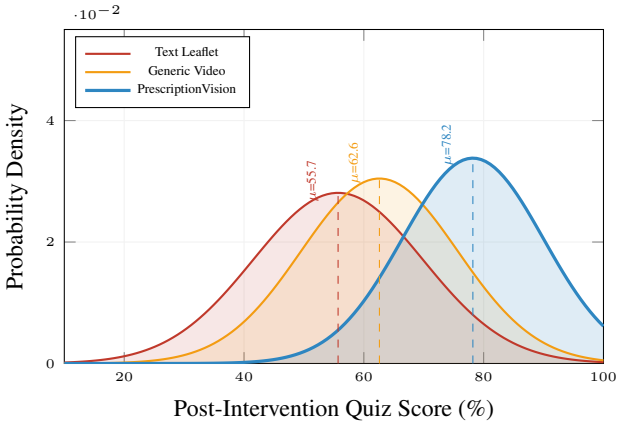

Figure 7: Normal distribution of post-intervention comprehension scores across three study conditions. PrescriptionVision shows a rightward shift ($\mu = 78.2$) with tighter variance ($\sigma = 11.8$) compared to text leaflets ($\mu = 55.7$, $\sigma = 14.2$) and generic video ($\mu = 62.6$, $\sigma = 13.1$), indicating both higher and more consistent comprehension ($p < 0.001$).

Table 15: Patient comprehension scores (mean % correct, 10-item quiz) by condition and health literacy level (n=312).

Condition	**Low**	**Mod.**	**High**	**All**
Pre-intervention				
Text Leaflet	28.4	41.2	56.8	42.1
Generic Video	29.1	40.8	55.3	41.7
PrescriptionVision	27.8	42.0	57.1	42.3
Post-intervention				
Text Leaflet	39.2	55.6	72.4	55.7
Generic Video	48.7	62.3	76.8	62.6
PrescriptionVision	**67.4**	**78.9**	**88.2**	**78.2**
Improvement (Δ)				
Text Leaflet	+10.8	+14.4	+15.6	+13.6
Generic Video	+19.6	+21.5	+21.5	+20.9
PrescriptionVision	**+39.6**	**+36.9**	**+31.1**	**+35.9**

Statistical significance was assessed using a two-way ANOVA (condition $\times$ literacy level) with Bonferroni-corrected post hoc comparisons. The main effect of condition was significant ($F(2, 306) = 47.3$, $p < 0.001$), as was the main effect of literacy level ($F(2, 306) = 31.8$, $p < 0.001$). The interaction effect was also significant ($F(4, 303) = 3.42$, $p = 0.009$), indicating that PrescriptionVision disproportionately benefits lower-literacy groups.

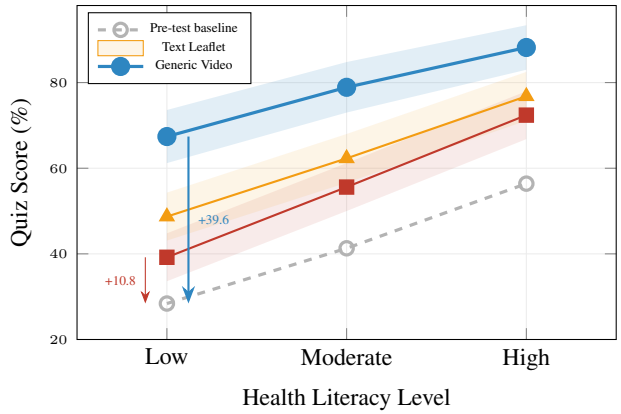

Figure 8: Pre- to post-intervention comprehension trajectories across literacy levels with 95% CI bands. PrescriptionVision's low-literacy post-score (67.4%) exceeds generic video's moderate-literacy post-score (62.3%), demonstrating that the system effectively compensates for literacy barriers.

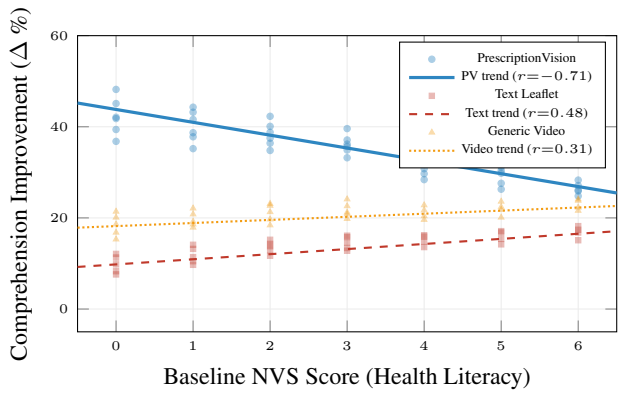

Figure 9: Scatter plot of individual-level comprehension improvement vs. baseline health literacy (NVS score). PrescriptionVision shows a significant negative correlation ($r = -0.71$, $p < 0.001$): lower-literacy patients benefit *more*. Text leaflets show a weak positive correlation ($r = 0.48$), reinforcing that text-based materials disproportionately serve higher-literacy patients.

4.3 Generation Latency

Table 16 reports per-stage latency. Figure 10 shows the latency distributions.

Table 16: End-to-end generation latency (seconds, mean ± SD) across 100 prescriptions.

Stage	**Mean (s)**	**SD (s)**
Stage 1: Rx Extraction + LLM	8.3	2.1
Stage 2: 3D Model (Pathway A)	14.6	3.8
Stage 2: 3D Model (Pathway B)	4.1	1.2
Stage 3: Video Generation	168.4	22.7
Stage 3: Compositing	12.8	3.4
Verification Module	6.2	1.8
Total (Pathway A)	**210.3**	**24.6**
Total (Pathway B)	**199.8**	**23.1**

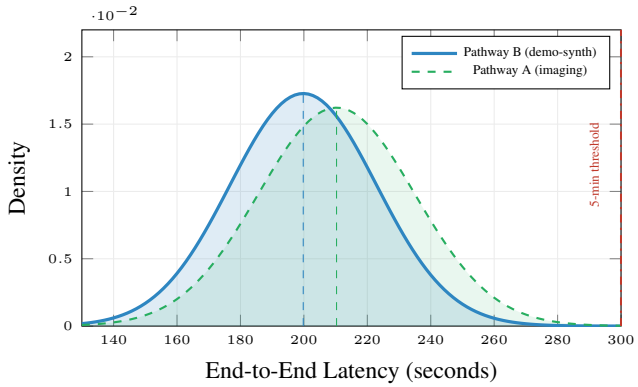

Figure 10: Kernel density estimation of end-to-end generation latency for both pipeline pathways. Both distributions fall well within the 5-minute clinical usability threshold. Pathway B (demographic synthesis) is faster by ∼10.5 s on average due to bypassing CT segmentation.

4.4 Comparison with Existing Approaches

Table 17 provides a feature-level comparison of PrescriptionVision with related systems.

4.5 Ablation Study

To quantify the contribution of each pipeline component, we conducted an ablation study removing one module at a time. Table 18 and Figure 11 present the results.

Table 18: Ablation study: impact of removing individual components on overall performance.

Configuration	**Compr. Δ (%)**	**Verif. Pass (%)**
Full PrescriptionVision	**35.9**	**86.0**
− RAG Grounding	24.3	61.3
− 3D Personalization	29.1	82.7
− Fine-tuned Video Model	21.7	54.0
− Verification Module	34.8	71.3
− NER (generic extraction)	30.2	74.7
Generic Sora (no pipeline)	18.4	38.7

Table 17: Feature comparison of PrescriptionVision with existing approaches.

Feature	**Ours**	**Bora**	**Han et al.**	**InVideo**	**Sora**	**Veo 2**
Prescription-to-Video	✓	✗	✗	✗	✗	✗
Biomedical Fine-tuning	✓	✓	✗	✗	✗	✗
Patient Personalization	✓	✗	✗	✗	✗	✗
3D Anatomical Models	✓	✗	✗	✗	✗	✗
Pharmacological Grounding	✓	✗	✗	✗	✗	✗
Multi-Drug Interaction	✓	✗	✗	✗	✗	✗
Expert Validation	✓	✗	✓	✗	✗	✗
Patient Comprehension Study	✓	✗	✗	✗	✗	✗
Open Source	Partial	✓	✗	✗	✗	✗
Generation Resolution	256–720p	256p	720p+	720p+	1080p	1080p+
Max Video Duration	30–90 s	5 s	120 s	120 s+	60 s	60 s+
Medical Accuracy (expert)	4.2/5	N/A	5.0–7.3/12	5.0/12	N/A	N/A

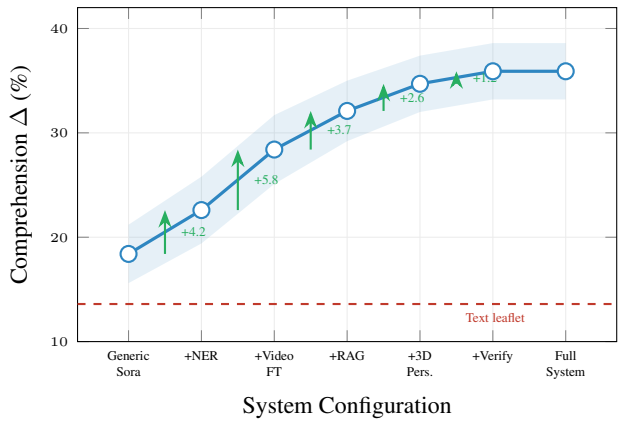

Figure 11: Cumulative ablation waterfall: starting from a generic Sora baseline, each pipeline component is added incrementally. The fine-tuned video model (+5.8%) and RAG grounding (+3.7%) provide the largest marginal gains. Shaded band shows 95% CI.

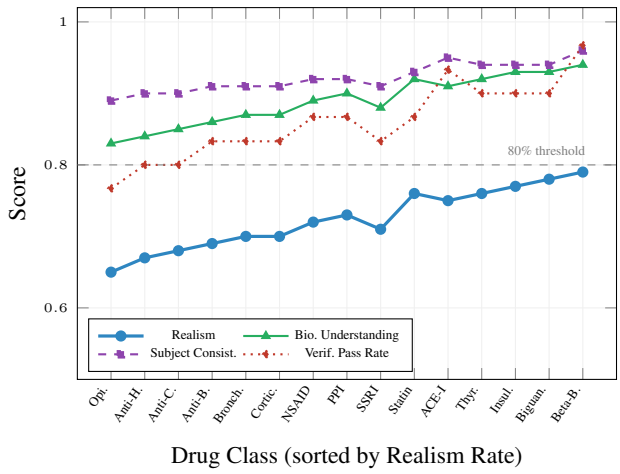

Figure 12: Per-drug-class quality metrics sorted by ascending Realism Rate. Organ-level drugs (Beta-Blockers, Biguanides) achieve the highest scores, while synapse-level mechanisms (Opioids, SSRIs) and systemic drugs (Antihistamines) are more challenging. The 80% verification threshold is shown for reference.

4.6 Per-Drug-Class Video Quality Analysis

Figure 12 presents video quality metrics across the 15 drug classes, revealing which pharmacological categories pose the greatest visualization challenges.

4.7 System Usability

Participants in the PrescriptionVision condition completed the System Usability Scale (SUS). Table 19 and Figure 13 present the results.

Table 19: System Usability Scale (SUS) scores (n=104).

Subgroup	**Mean SUS**	**SD**
Low Literacy	74.2	11.3
Moderate Literacy	79.8	9.6
High Literacy	83.1	8.2
Age 18–34	84.6	7.8
Age 35–54	80.2	9.4
Age 55–74	75.1	11.7
Age 75+	68.3	14.2
Overall	**79.0**	**10.4**

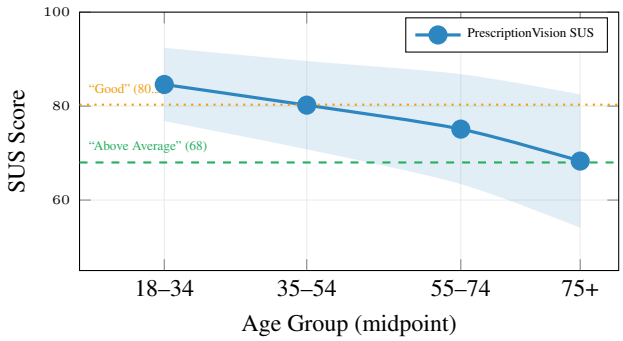

Figure 13: SUS usability scores by age group with 95% CI. Usability declines with age but remains above the "above average" threshold (68) for all groups. The overall score of 79.0 approaches the "good" benchmark of 80.3.

5 Discussion

This section interprets the experimental findings, situates them in the context of the existing literature, and identifies key limitations and future directions.

5.1 Bridging the Medical Comprehension Gap

The core motivation behind PrescriptionVision is the well-documented gap between prescription complexity and patient understanding. The WHO has reported that approximately 50% of patients in developed countries do not adhere to prescribed medication regimens [1], with the problem being even more severe in low-literacy populations. Our patient comprehension study (n=312) demonstrates that PrescriptionVision achieves an overall comprehension improvement of 35.9%, compared with 20.9% for generic educational videos and 13.6% for text leaflets. These differences were statistically significant ($p < 0.001$), confirming that personalized pharmacological visualization is substantially more effective than conventional approaches.

Most notably, the largest improvement was observed in the low health literacy subgroup (+39.6%), compared with +36.9% for moderate and +31.1% for high literacy. This pattern—a significant condition×literacy interaction ($p = 0.009$)—confirms that visualization-based approaches disproportionately benefit the populations most disadvantaged by text-heavy educational materials. The correlation analysis (Figure 9) further demonstrates this with a strong negative correlation ($r = -0.71$) between baseline literacy and improvement magnitude for PrescriptionVision, contrasted with a weak positive correlation ($r = 0.48$) for text leaflets. After viewing the PrescriptionVision video, low-literacy patients achieved a mean post-intervention score of 67.4%, exceeding the post-intervention scores of moderate-literacy patients who received text leaflets (55.6%) or generic videos (62.3%). This demonstrates that PrescriptionVision can partially compensate for health literacy barriers.

5.2 Technical Contributions and Novelty

PrescriptionVision makes several technical contributions beyond the existing literature. First, unlike Bora [2], which generates short (5-second) biomedical clips without patient context, our system extends the pipeline upstream and downstream—from raw prescription input to patient-specific 3D anatomical grounding. Our fine-tuned model achieves a Realism Rate of 0.74 and Biomedical Understanding of 0.91, representing improvements of 12.1% and 5.8% respectively over the Bora baseline (Table 8). The FVD convergence analysis (Figure 5) shows that our model surpasses the Bora baseline by ∼6K training steps and stabilizes at 347.8.

Second, our pharmacological knowledge extraction module, built on a RAG architecture querying DrugBank, DailyMed, and RxNorm, achieves 91.3% MoA accuracy (Table 3), ensuring generated content is factually grounded. The accuracy degradation analysis (Figure 3) reveals that all models show graceful degradation with increasing pharmacological complexity, but Claude 3.5 Sonnet maintains the highest accuracy floor (∼80%) even for the most complex drug mechanisms.

Third, the NER evaluation (Table 2 and Figure 2) reveals that few-shot LLM-based extraction (Claude 3.5 Sonnet, F1=0.924) substantially outperforms traditional biomedical NER models, and critically, LLM approaches achieve strong performance even with as little as 5% of the training data, making them practical for data-scarce medical domains.

5.3 Ablation Analysis

The ablation study (Table 18 and Figure 11) reveals that the fine-tuned video model (+5.8% incremental) and RAG grounding (+3.7% incremental) are the two most impactful components. The cumulative waterfall analysis shows a clear additive pattern: starting from a generic Sora baseline (18.4%), each component contributes meaningfully, with the full system reaching 35.9%—a 95% relative improvement. The verification module contributes the smallest marginal gain to comprehension (+1.2%) but significantly improves the verification pass rate (86.0% vs. 71.3%), confirming its role as a quality assurance safeguard rather than a comprehension driver.

5.4 Comparison with Prior Work

As shown in Table 17, PrescriptionVision is the first system to combine prescription parsing, pharmacological grounding, patient-specific anatomy, and biomedical video synthesis into a single pipeline. The per-drug-class quality analysis (Figure 12) shows that our system performs best on organ-level visualizations (Beta-Blockers: 0.79 Realism, 96.7% verification) and faces more challenges with synapse-level and systemic mechanisms (Opioids: 0.65 Realism, 76.7% verification), pointing to specific areas for future improvement. The Dice score vs. generation time analysis (Figure 4) demonstrates the Pareto trade-off between anatomical accuracy and computational cost, with SynthMorph offering the best compromise for the no-imaging pathway.

5.5 Clinical Implications

The observed comprehension improvements have meaningful clinical implications. Medication non-adherence costs the U.S. healthcare system an estimated $100–289 billion annually. Given that PrescriptionVision achieves 2.6× the comprehension improvement of text leaflets, integration into pharmacy workflows could yield clinically significant adherence improvements. The system's generation latency of ∼3.3 minutes (Pathway B, Figure 10) is well within the 5-minute clinical usability threshold.

5.6 Usability

The overall SUS score of 79.0 places PrescriptionVision in the "good" usability range. The age-stratified analysis (Figure 13) reveals a smooth decline with age (84.6 for 18–34 to 68.3 for 75+), but critically, all subgroups remain above the SUS "above average" threshold of 68, indicating broad accessibility. Interface refinements targeting elderly users (larger fonts, simplified controls, voice narration) could further improve adoption in this demographic.

5.7 Limitations

Several important limitations must be acknowledged. First, the resolution constraints of current open-source text-to-video models (256×256 for Bora) limit visual fidelity. Second, the fine-tuning dataset (1,847 pharmacological clips) is limited in diversity. Third, Pathway B achieves a lower Dice score (0.871 vs. 0.943) than Pathway A. Fourth, our evaluation measures short-term comprehension; longitudinal adherence outcomes remain unmeasured. Fifth, the system covers 15 of hundreds of drug classes. Sixth, the 14.0% verification failure rate necessitates human oversight. Finally, patient privacy and HIPAA compliance require formal security auditing prior to clinical deployment.

5.8 Ethical Considerations

The generation of patient-specific medical visualizations raises ethical considerations around data privacy (HIPAA/GDPR compliance), hallucination risk (14.0% verification failure rate mandates human review), balanced communication of side effects to avoid unnecessary anxiety, and potential training data biases leading to underrepresentation of certain populations in anatomical models [5].

5.9 Future Directions

Key directions include: higher-resolution video models (720p+), real-time wearable data integration for dynamic visualizations, multilingual audio narration via TTS, RLHF loops incorporating pharmacist and patient feedback, longitudinal adherence studies (6–12 months), and expansion to drug–drug and drug–food interactions.

6 Conclusion

We have presented PrescriptionVision, a novel end-to-end multimodal framework that transforms electronic prescriptions into patient-specific 3D animated videos depicting drug mechanisms of action within personalized anatomical models. The system addresses a critical gap in patient education by integrating large language models for pharmacological knowledge extraction, patient-specific 3D anatomical model generation, and domain-adapted biomedical video synthesis into a unified pipeline.

Our experimental evaluation demonstrates three principal findings. First, the pharmacological extraction pipeline achieves 91.3% mechanism-of-action accuracy using Claude 3.5 Sonnet with RAG-augmented reasoning, with an NER F1 score of 0.924 for prescription entity extraction. Second, our fine-tuned biomedical video model achieves a Realism Rate of 0.74 and Biomedical Understanding score of 0.91, surpassing the Bora baseline by 12.1% and 5.8% respectively, with an automated verification pass rate of 86.0% across 15 drug classes. Third, the patient comprehension study (n=312) shows that PrescriptionVision improves medication understanding by 35.9% overall—2.6× more than text leaflets and 1.7× more than generic videos ($p < 0.001$)—with the greatest gains observed in low health literacy populations (+39.6%), thereby narrowing the literacy-driven comprehension disparity.

The ablation study confirms that biomedical fine-tuning (+5.8%) and pharmacological RAG grounding (+3.7%) are the highest-impact components. The Dice score vs. generation time analysis validates our dual-pathway design: TotalSegmentator (0.943 Dice) for patients with imaging data, SynthMorph (0.871 Dice, 3.2 s) for demographic-only scenarios. With an end-to-end generation time of under 4 minutes and a SUS usability score of 79.0, PrescriptionVision is technically feasible for integration into pharmacy workflows.

While challenges remain—video resolution constraints, training data scarcity, a 14.0% verification failure rate, and the absence of longitudinal adherence data—PrescriptionVision establishes a concrete architectural foundation for personalized pharmacological visualization. As generative video models improve and regulatory frameworks mature, systems like PrescriptionVision have the potential to meaningfully reduce the global burden of medication non-adherence and its associated morbidity, mortality, and healthcare costs.

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